Johnson Yep

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Education

University of Waterloo — Waterloo, Ontario
Mechatronics and Robotics Engineering — April 2030
Relevant Courses: OOP in C++, Data Structures and Algorithms, Linear Algebra, Circuits

Skills

- **Languages:** Python, C++, TypeScript, Java
- **Perception & ML:** OpenCV, PyTorch, Ultralytics YOLO, Roboflow, MediaPipe, Reinforcement Learning, Point Clouds, Hand-Eye Calibration, Homography
- **Robotics:** ROS 2, MoveIt, Isaac Lab, Isaac Sim, Foxglove, Universal Robots UR20, AMRs
- **Web & Dev Tools:** Next.js, React, Jest, Git, Docker, Linux, GitHub Actions
- **Controls & CAD:** Allen-Bradley PLCs, Ladder Logic, EtherNet/IP, HMI, SolidWorks, AutoCAD, GD&T

Experience

Robotics & Software Engineering Intern — Sept 2026 - Present
RoBIM Technologies — View Work

- Developed vision software for RoBIM's robot cells, which prefabricate walls for construction, and prototyped it on a UR20 arm with a wrist-mounted camera to inspect and sort lumber.
- Fine-tuned a **YOLO** instance-segmentation model to detect boards in the camera feed, trained on a custom dataset labelled in **Roboflow**; reached 0.836 mask mAP50-95 on held-out images.
- Used this model to build a lumber inspection pipeline with **ROS 2**, **MoveIt**, **Python**, and **C++** that locates a board anywhere in the work area and scans it into a **3D point cloud** that captures warp and defects; measured board dimensions within 5 mm of a tape measure across 50+ boards.
- Reused the pipeline's board localization to develop a pick-and-place system that sorts boards by length, and designed its gripper attachment in **SolidWorks**.
- Built the operator web app with **Next.js**, **React**, and **TypeScript** to run both scans and sorts, with a live camera feed showing YOLO masks, board selection, robot controls, and sort progress; tested with **Jest**.

Robotics Engineering Intern — Jan 2026 - May 2026
Toyota Boshoku — View Work

- Automated 100+ rack moves per shift by setting up a **MiR AMR** to tow racks fitted with hook bars and **AprilTags**, mapping its routes and pickup and drop-off points, and programming its mission logic.
- Let operators queue AMR missions from wireless call buttons by linking a wireless controller to an **Allen-Bradley PLC** with **ladder logic** that queues requests and starts them through the **MiR API**.
- Built a **computer vision** system in **Python** and **OpenCV** that detects open drop-off spots from an overhead camera using lens correction, **homography** and HSV thresholds.
- Saved 30+ seconds per drop-off by sending those spots to the PLC over **EtherNet/IP** so the AMR drives straight to them, falling back to lane sensors if the vision heartbeat drops.

Humanoid Robotics Team — May 2026 - Present
WATonomous — View Work

- Trained a **PPO** policy in **Isaac Lab** for an SO-ARM101 arm to push objects through contact-only manipulation, using **11** shaped reward terms and a performance-gated curriculum to reach **85%** success on fully randomized spawns.
- Built a **32×16**-taxel open-source tactile sensor with a real-time visualization pipeline.

Driveline Team — Sep 2025 - May 2026
uWaterloo Formula Electric — View Work

- Designed and manufactured drilling and motor-mounting jigs in **SolidWorks** with **GD&T** drawings, machining features to **±0.1 mm**.

Projects

Autonomous Checkers Robot *C++* — View Project

- Built an autonomous robot that detects board pieces, selects moves, and physically moves the pieces.

Hand Tracking Computer Vision Guitar *Python, OpenCV, MediaPipe* — View Project

- Converted real-time hand and finger tracking into playable chords and notes.

Space Race *Java, CSS* — View Project

- Developed a 2D shooter game from scratch, designing gameplay, UI, and architecture end-to-end.